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### Surgical saw system including a reciprocating saw blade cartridge

#### Abstract

A surgical saw system including a saw, a blade cartridge and an attachment. The saw including a housing and a reciprocating shaft. The blade cartridge that may be removably coupled to the saw includes a bar and a rack. The bar includes a first holding feature. The rack is moveably disposed within the bar and is configured to removably couple to the reciprocating shaft of the saw. The attachment is disposed between and connecting the bar of the blade cartridge to the saw. The attachment may include a coupling feature in selective releasable engagement with the first holding feature of the bar to hold the bar static relative to the saw. The blade cartridge may also include a drive link that extends from reciprocating shaft of the saw and the rack for imparting the reciprocal motion of the reciprocating shaft to the rack.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This patent application is a continuation of U.S. patent application Ser. No. 17/478,327 filed Sep. 17, 2021, which is a continuation of U.S. patent application Ser. No. 16/062,727 filed Jun. 15, 2018, which is a national phase of International Patent Application No. PCT/US2016/066981 filed Dec. 15, 2016, which claims priority to and all advantages of U.S. Provisional Patent Application No. 62/268,536 filed Dec. 17, 2015, the entire contents of which are incorporated herein by reference.

## FIELD OF THE INVENTION

(1) This application is generally related to a reciprocating saw blade. More particularly, this application is directed a reciprocating saw blade cartridge that is releasably mounted to the saw with which the cartridge is used. The cartridge includes a static bar to which a reciprocating rack is attached.

## BACKGROUND OF THE INVENTION

(2) In a surgical procedure, it is sometimes necessary to use a saw to remove tissue, including bone and cartilage. Often a powered saw is used to perform this procedure. Attached to the saw is a saw blade. A drive assembly internal to the saw oscillates the blade in a back and forth motion. Some blades move back and forth along their longitudinal axes. This type of blade is known as a reciprocating saw blade. This type of blade is provided with teeth that extend outwardly from a side edge of the blade body. Many reciprocating saw blades are designed to remove hard tissue, bone.

(3) Reciprocating saw blades generally work well for the purposes for which they are designed. These saw blades are able to cut the bones to which they are applied.

(4) However, there are times when use of a reciprocating saw blade results in undesirable side effects. One such side effect is due to the fact that, when a reciprocating saw blade is positioned to cut some types of tissue, the body of the blade presses against soft tissue that is not targeted for removal. This can occur when a reciprocating saw blade is used in oral surgery to remove a portion of the jaw. The body of a reciprocating saw blade may also press tissue when the blade is used in small bone orthopedic surgery, surgery of the hand or foot and ankle.

(5) A problem can arise in these types of surgical procedure because a reciprocating saw blade can move back and forth at a rate of between 10,000 and 15,000 cycles per minute. This rapid movement of the blade can foster appreciable friction-induced heating of the soft tissue against which the blade body is pressed. The heating of this tissue can sometimes damage the tissue. If the damages tissue is part of the nerve system, a muscle or a ligament, the effects of the damage can especially undesirable.

(6) Also, there are procedures in which, owing to how the reciprocating saw blade is positioned in the tissue, it is extremely difficult, if not impossible, for the surgeon to see the end section of the blade. For example in a rhinoplasty procedure a reciprocating saw blade is used to remove a portion of the nose bone. In this type of procedure, to avoid the ready appearance of a scar, the blade is typically inserted into the nose through the flap of the skin between the nostrils. In some hip replacement procedures a reciprocating saw blade may likewise be positioned so it can be difficult to see the free end of the blade. Again, it should be understood that a reciprocating blade moves back and forth. In procedures in which the distal section of the blade is hidden from view, there is a possibility that this section of the blade repeatedly presses against tissue that is not targeted for removal. This back and forth movement of the tissue can stress the tissue to the point at

which it tears or is otherwise damaged.

(7) In some procedures, to ensure a planar cut, it is a common practice to insert the reciprocating saw blade in a device known as a resection guide (not illustrated). This guide is sometimes referred to as a guide block or a cutting jig. The resection guide has a slot in which the blade is inserted. The device is held static relative to the tissue to be cut. Since the device is held static, the blade is blocked from movement out of the plane of the slot. By so restricting the movement of the blade, the surface of the bone left after the cut is formed s planar in shape. This facilitates the proper seating of an implant, which typically has its own planar surface, against the bone. When a reciprocating saw blade moves back and forth in the resection guide, the body of the blade invariably rubs against the inner surfaces of the guide that define the cut. This metal-against-metal rubbing results in the rubbing away of the material forming the resection guide. One adverse effect of this wear is that at least some of the fine dust of metal generated by this wear invariably lands on the exposed internal tissue of the patient. This means that after the cut is formed and the resection guide is removed, care must be taken to clean the tissue to remove these debris so as to prevent them causing harm to the patient. A second adverse effect of this wear is that, over time, the wear can appreciably widen the slot or slots formed in the resection guide. If these slots become too wide relative to the blade, the resection guide may no longer function as a useful guide for ensuring that the cut formed by the reciprocating blade has the desired shape.

(8) US Pat. Pub. US 2006/0009796 A1/PCT Pub. WO 2006/017066 A2, the contents of which are incorporated herein by reference, discloses a type of a saw blade known as a sagittal saw blade. A sagittal saw blade has teeth that pivot back and forth around an axis that extends through the plane of the blade. One of the sagittal saw blades disclosed in this publication has a blade head that extends laterally outwardly from the side of a static blade bar. The blade head pivots around an axis that extends through the opposed major surfaces of the blade bar. This blade head, when pivoted, cuts the tissue against which the teeth are pressed. However, the length of the cut formed by this blade is relatively short, typically less than one-third length of the cut that would be performed by a compatibly sized conventional reciprocating saw blade. It therefore can take an appreciably longer time to make a cut with this cartridge than it would in order to make the same length cut using a conventional reciprocating saw blade.

(9) A species of the reciprocating saw blade is the sternum saw. As implied by its name, the sternum saw blade is used to cut the sternum. Typically, it is necessary, to cut, separate, the sternum so the surgeon can gain access to the tissue and organs below the sternum. These organs include the heart and lungs. When cutting the sternum, it is naturally desirable to avoid positioning the saw blade so the blade strikes, cuts, the organs and tissue immediately below the blade. To prevent this undesirable result, a sternum saw blade is mounted in a guard. A foot extends outwardly from the free end of this guard. This foot, in comparison to a narrow width blade, is relatively wide. During the process cutting the sternum, the surgeon holds the saw so the foot of the sternum guard presses against the undersurface of the sternum. The foot functions as a stop that ensures that, as the blade is advanced along the sternum, the blade cuts completely through the sternum from the outer surface to the undersurface. Further, by ensuring that the blade is so positioned, the guard ensures that the blade does not become so positioned that, as the blade is advanced the blade cuts the organs and tissues that underlying the sternum.

(10) Sternum guards work well for the purpose for which they are designed. A problem with some sternum guards is that is that they are designed to be used with saws especially designed to hold these guards. This means a surgical facility in which the sternum is to be cut is required to have one or more of the special saws, called sternum saws, available. The facility may be required to have these saws available even though these saws have essentially no use outside of performing this one specific procedure.

## SUMMARY OF THE INVENTION

(11) This disclosure describes a new and useful reciprocating saw blade. The reciprocating saw

blade is constructed to minimize the extent to which moving components of the blade other than the teeth, press against tissue. Some versions of the saw blade are further constructed to be used as a sternum blade without requiring the use of a supplemental sternum guard.

(12) The saw blade is referred to as a cartridge. The cartridge includes an elongated bar. The bar includes features that facilitate the releasable attachment of the bar to the saw used to actuate the blade. One side of the bar is open.

(13) The cartridge also includes a rack. The rack has a base that is moveably disposed in the bar. A portion of the rack extends to the opening in the side of the bar. The rack also includes a set of teeth. The teeth extend outwardly from the section of the bar adjacent the opening in bar. The teeth are wider in depth than the base of the rack. More particularly, have sufficient width so that when the rack is reciprocated and the teeth pressed against tissue, the teeth form a kerf of sufficient that the cartridge is able to advance through the kerf.

(14) The rack and bar may be formed with complementary features that retain the rack in the bar while allowing the rack to move linearly in the bar.

(15) Some versions of the cartridge may be formed to have a foot that protrudes outwardly from the opposed major surfaces of the bar. This foot functions in a manner similar to the foot of a sternum guard. Thus the foot functions as a stop to facilitate the proper positioning of the cartridge relative to the bone the cartridge is employed to cut.

(16) This application is also directed to an attachment that can be used to couple the above-described blade cartridge to a conventional saw. A conventional saw is understood to be a saw that can be used to reciprocate a conventional reciprocating saw blade. The attachment includes a body. The body includes features for releasably holding the body static to the distal end of a conventional saw. Attached to the body is a coupling feature. The coupling feature is shaped to engage with a complementary coupling feature integral with the bar of the cartridge so as to releasably hold the bar static to the attachment. The attachment coupling feature may be a member dimensioned to seat in a void formed in the bar. The coupling feature may alternatively be an opening or other void in the body that receives the complementary coupling feature integral with the bar.

(17) This application also discloses a saw specifically for use with the disclosed blade cartridge. This saw has a feature for engaging the bar of the cartridge to releasably hold the bar static to the body of the saw.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following Detailed Description is to be read with the accompanying drawings in which:
- (2) FIG. 1 is side view of a reciprocating saw to which a reciprocating blade cartridge is attached;
- (3) FIG. 2 is a top view of how a reciprocating blade cartridge is attached to the associated saw;
- (4) FIG. 3 is an exploded view of the grasping assembly that holds the reciprocating blade cartridge to the saw;
- (5) FIG. 4 is a perspective view of the reciprocating blade cartridge;
- (6) FIG. 5 is an exploded view of the reciprocating blade cartridge;
- (7) FIG. 6 is a cross sectional view of the reciprocating blade cartridge;
- (8) FIG. 7 is a side view of a second reciprocating saw and the complementary second reciprocating blade cartridge that is used with the saw;
- (9) FIG. 8 is a top view of the saw and blade cartridge of FIG. 7;
- (10) FIG. 9 is an exploded view of the saw, front end attachment and blade cartridge of FIG. 7;
- (11) FIG. 10 is a cross sectional view of the blade cartridge of FIG. 7 coupled to the complementary front end attachment;
- (12) FIG. 11 is an exploded view of the reciprocating blade cartridge of FIG. 7;

- (13) FIG. 12 is a side view of a third reciprocating blade cartridge, the cartridge being coupled to a saw;
- (14) FIG. 13 is an exploded view of the basic components of the saw, front end attachment and reciprocating blade cartridge of FIG. 12;
- (15) FIG. 14 is a cross sectional view of the blade cartridge of FIG. 11 coupled to the complementary saw attachment;
- (16) FIG. 15 is an exploded view of the reciprocating blade cartridge of FIG. 11 and the associated front end attachment;
- (17) FIG. 16 is a plan view of a third reciprocating blade cartridge; and
- (18) FIG. 17 is a cross sectional view taken along line 17-17 of FIG. 16 of the blade cartridge of FIG. 16.

## DETAILED DESCRIPTION

### I. First Embodiment

- (19) FIG. 1 illustrates a surgical reciprocating saw 30 to which a complementary reciprocating blade cartridge 80 is removably attached. The saw 30 has a pistol shaped body or housing 32. The saw housing 32 includes a grip 34. A barrel 36 is formed integrally with handgrip 34. The barrel 36 extends proximally, rearwardly, from and distally forward of the handgrip 34. (Here, “proximally” is understood to mean towards the surgeon holding the saw 30, away from the surgical site to which the blade cartridge 80 is applied. “Distally” is understood to mean away from the surgeon and towards the site to which the blade cartridge 80 is applied.) A nose 38 projects forward from the barrel 36 at a distal end of the housing 32.
- (20) Internal to the saw barrel 36 is a motor, represented by a dashed rectangle 40 and a reciprocating shaft, represented by dashed rectangle 46. A transmission, represented by dashed rectangle 44, is located between the motor 40 and reciprocating shaft 46. The transmission converts the rotational movement of the motor rotor into motion that reciprocates the shaft back and forth. Here “reciprocation” is understood to mean a repetitive proximal-to-distal-to-proximal motion along the longitudinal axis of the component under discussion. The distal end of the reciprocating shaft 46 is provided with engagement features that facilitate the removable coupling of the below discussed cartridge rack 112 to the shaft. The exact structure of these engagement features are not part of the invention, and may be readily provided by one skilled in the art. By way of an example, the shaft 46 may be provided with openings designed to receive holding features, e.g., complementary components, integral with the rack. A chuck, represented by dashed ring 47, may be fitted to the distal end of the shaft. The chuck 47 may hold the holding features integral with the rack 112 to the shaft.
- (21) The motor 40 internal to the saw may be a DC motor. A battery (not illustrated) may be attached to the base of the saw handgrip 34. A trigger 48 extends forward from the distally directed face of the handgrip 34. Internal to the handgrip is a control module represented by dashed rectangle 50. The control module 50 is configured to control the sourcing of the current from the battery or other power supply to the motor 40. The control module also includes a transducer that monitors the actuation of the trigger 48. Based on the actuation of the trigger 48, the control module 50 regulates the application of current to the motor 40 so as to regulate the actuation of the saw 30 and by extension, the actuation of the blade cartridge 80.
- (22) Many versions of saw 30 are configured to reciprocate a conventional reciprocating saw blade (not illustrated). The chuck 47 may be designed to hold a conventional, i.e. prior art, reciprocating saw blade (not illustrated) to reciprocating shaft 46 so the blade can be reciprocated by the shaft 46.
- (23) An attachment 52, seen best in FIG. 2 and FIG. 3, is removably secured to the saw nose 38. The attachment 52, together with the saw 30 and the blade cartridge 80, may collectively comprise a first example saw system. The attachment 52 releasably holds the below described bar 82 of the blade cartridge 80 to the saw 30. Attachment 52 includes two shells 54. Shells 54 form the body of the attachment 52. Each shell 54 includes a planar base 56 and two wings 58 that extend outwardly

from opposed sides of the base **56**. Each shell **54** is shaped so that as the wings **58** extends away from the shell base, the wings curve away from the base. When the saw **30** is assembled, shells **54** are placed on the opposed sides of the nose **38** so the ends of the wings **58** of each shell **54** face the ends of the wings of the other shell. Bolts **60** extends through each wing **58** to the opposed adjacent wings. Bolts **60** clamp the shells **54** together, the bolts and shells collectively defining a clamping assembly, i.e., an example first coupling feature, to hold the shells **54** around the opposed sides of the nose **38**. Shells **54** and bolts **60** thus releasably hold the attachment **52** static to the nose **38**.

(24) A hand **64**, also part of the attachment **52**, extends distally forward from each shell **54**. Each hand **64** includes a rectangular palm **66**, one palm identified. Palms **66** are dimensioned to seat over the exposed surface of the shell **54** from which the hand **64** extends. Screws **74** hold each palm **66** to the associated shell **54**. (Not identified with reference numbers are the openings in the shells **54** and hands **64** through which the screws **74** extend. A finger **68**, also part of each hand **64**, extends forward from each palm **66**. Each finger **68** has a main section (not identified) that extends along a longitudinal axis that is parallel to the proximal-to-distal longitudinal axis through the saw barrel **36**. Each finger **68** has a tip **70** that extends perpendicularly to the longitudinal axis of the main section of the finger. A small rib **72**, an example second coupling feature, protrudes forward from end face of each finger tip **70**. The clamping assembly **58** and **60** is arranged so that when the clamping assembly is fitted to the saw, the opposed ribs **72** face each other.

(25) From FIGS. **4** and **5** it can be seen that the reciprocating blade cartridge **80** includes a static bar **82** in which a rack **112** is moveably mounted. Bar **82** includes two plates **84**. The plates **84** are generally rectangular in shape. Each plate **84** is further formed so as to have between the distally directed face **86** and the top face **90** a transition face **88**. In FIG. **5** the edges of faces **86**, **88**, and **90** are identified. Each transition face **88** is tapered in that the face **88** angles both upwardly as the face extends proximally away from the associated distally directed face **86**. Each plate **84** may be formed with plural slots. A first slot, slot **92**, an example first holding feature, for removably holding the bar **82** static to the saw **30**, is formed in the bar so as to be forward of the proximal end of the plate **84**. The plates **84** are formed so that slots **92** extend along axes that are perpendicular to the proximal-to-distal longitudinal axes through the plates. Each slot **92** is dimensioned to receive a separate one of the ribs **72** integral with the finger tips **70** and the ribs **72** are likewise dimensioned to seat in the slots **92**. The second slot, i.e., a debris slot **94**, formed in each plate **84** is located immediately proximal to the distal and transition faces **88** and **90**, of the plates **84**. Each slot **94** thus has a first section that adjacent the plate distal face that is parallel to slot **92** and a section that extends diagonally away from the first section. The second section of each slot **94** extends parallel to the transition face **88** of the plate **84** in which the slot **94** is formed.

(26) The cartridge bar **82** also includes a spacer **98** that is sandwiched between the plates **84**. Spacer **98** includes an elongated beam **102**. The beam **102** has three sections, individual sections not identified with reference numbers. A first section of the beam **102** extends distally from the proximal end of the plates **92** so as to be flush with the top faces **90** of the plates. A second section of the beam **102**, as it extends forward from the first section, angles downwardly so as to be flush with the transition faces **88** of the plates **84**. The third section of the beam **102** extends downwardly from the second section so as to be flush with the distal faces **86** of the plates **84**.

(27) Spacer **98** also has two webs **104**, one web being identified in FIGS. **5** and **6** with reference number **104**. Webs **104** extend perpendicularly away from the first section of the beam **102**. Webs **104** are parallel. The foot **106**, one foot identified also part of spacer **98**, is located at the end of each web **104**. Feet **106** are parallel to the first section of the beam **102**. Each foot **106** extends both proximally away from and distally away from the web **104** with which the foot is integral.

(28) When the cartridge **80** is assembled, spacer **98** is welded between the plates **84**. In FIG. **4** the weld lines **108** formed by the penetration welding of one plate **84** to the underlying webs **104** and feet **106** are seen.

(29) Rack **112** includes a base **118**. The base **118** is dimensioned to move in the space within the

bar **82** between the plates **84** and around the web **104** and feet **106** of the spacer **98**. Not identified with reference numbers are the top and distally directed surfaces of the rack base **118**. Likewise not identified is the transition surface between the top and distally directed surfaces. The rack base **118** is formed with two T-shaped openings, i.e., guide slots, **120**. Each opening **120** has a center section that is wider in width than the proximal to distal width across the webs **104**. The end section of each opening **120** is longer in length than the length of the spacer feet **106**. When the cartridge **80** is assembled, each spacer web **104** and associated foot **106** seats in one of the openings **120** in the rack base **118**. Owing to the dimensioning of the components forming the cartridge, it should be understood that the rack **112** is able to move longitudinally back and forth in the bar **82**.

(30) A tail **114**, also part of the rack **112**, extends proximally from the base **118** so as to extend out of the bar **82**. The tail has features, i.e., second holding features, that facilitate the removable coupling of the tail to the reciprocating shaft **46** internal to the saw **30**. The tail may be planar in shape, and has a top to bottom height, less than the same dimension of the base. Further, the tail may be formed so that two tabs **116** extend from the opposed top and bottom sides of the tail. Tabs **116** are diametrically opposed to each other relative to the proximal to distal longitudinal axis through the tail **114**.

(31) Rack **112** also has a head **124** that extends outwardly from the base **118**. The head **124** extends outwardly from the side of the base opposite the side of the base in which openings **120** are formed. The rack **112** is formed so that when the cartridge **80** is assembled, head **124** is located outside of and immediately adjacent the bottom of the bar, the portion of the rack opposite the plate top faces **90**. The head **124** is formed so as to have teeth **126**. The teeth **126** extend linearly, proximally to distally, along the head. The head and, more particularly the teeth have a side-to-side width that is greater than the thickness between the opposed major faces of the base of the rack **112**. More specifically, the teeth have a width such that the kerf cut by the teeth can accommodate cartridge bar **82**. This means that the teeth have a side to side width that is at least 0.2 mm greater than the side to side thickness across bar **82**.

(32) This saw system is prepared for use by coupling the cartridge **80** to the saw **30**. An initial step of this process is the coupling of the rack tail **114** to the reciprocating shaft **46** internal to the saw **30**. Tail **114** thus functions as the drive link that includes the second holding features and connects the reciprocating shaft **46** of the saw **30** to the rest of the rack **112** as may be done with the chuck **47**. The clamp assembly hands **64** are secured over the cartridge so that the ribs **72** integral with the arms seat in slots **92** formed in the cartridge bar **82**. This completes the removable attachment of the cartridge **80** to the saw **30**. The saw **30** and cartridge **80** are ready for use.

(33) The saw **30** and cartridge **80** are used by positioning the cartridge teeth **126** adjacent the tissue, typically bone, in which the cut is to be formed. The trigger **48** is depressed. The control module **50**, in response to the detecting that depression of the trigger **18**, actuates the motor **40**. The actuation of the motor **40** results in the reciprocation of the reciprocating shaft **46** internal to the saw **30**. The movement of the shaft **46** causes a like linear reciprocal movement of the rack **112**. It should be appreciated that, as the rack **112** reciprocates, feet **106** internal to the bar **82**, hold the rack **112** in the bar **82**. The teeth **126** are pressed against the tissue to be cut. The reciprocation of the teeth **126** when pressed against the tissue result in the teeth forming the desired cut in the tissue.

(34) When the saw **30** is used to cut tissue, debris formed in the cutting process enter the space between the bar plates **94**. The movement of the rack **112** forces the debris out of the bar through slots **94**. This prevents the debris from clogging the inside of the bar **82** to such an extent that the debris impede the reciprocation of the rack **112**.

(35) The cartridge **80** is designed so that when actuated, the only exposed portion that moves is the rack head **124**. The sides of bar **82** that are pressed against tissue do not reciprocate. Since the bar **82** does not reciprocate, the possibility that the tissue could be damages such motion is eliminated.

(36) The fact that the cartridge bar **82** does not reciprocate also means that when the distal end of the cartridge **80** is pressed against tissue, this tissue is not exposed to back and forth motion as a



result of the actuation of the cartridge **80**. Since this tissue is not exposed to this motion, it is not subjected to the stress of this motion that can tear or otherwise damage the tissue.

(37) Furthermore, when the cartridge **80** is inserted in the slot of the above-discussed resection guide and actuated, the cartridge bar **82** does not reciprocate. This eliminates the wear to that otherwise occurs as a result of the metal-against-metal reciprocal movement of the whole of the blade in the resection guide. The elimination of this wear results in a like elimination of the adverse effects of this wear.

(38) The webs **104** and feet **106** may extend through the void space in the bar **82** between the opposed inner surfaces of plates **84**. Webs **104** and feet **106** thus provide the bar **82** with stiffness. The feet **106** thus perform two functions, retaining the rack **112** in the bar **82**, and also providing structural strength to the bar **82**.

(39) It should likewise be understood that attachment **52** allows cartridge **80** to be used with a conventional saw **30**. This means that a facility that wants to use cartridge **80** is not required to purchase a saw **30** the only purpose of which is to be used with cartridge **80**.

## II. Second Embodiment

(40) FIGS. 7-9 illustrate the basic features of an alternative reciprocating saw **130** and an alternative blade cartridge **180**. Saw **130** has a tubular or pencil shaped body **132**. A motor, represented by a dashed rectangle **134** is disposed in the body **132**. A reciprocating shaft, represented by dashed rectangle **140**, extends forward of the motor. A transmission, not depicted, is located between the motor **134** and the reciprocating shaft **140**. The transmission converts the rotational movement of the motor rotor into motion that reciprocates the shaft back and forth. Not shown are the components that power the motor **134**. If the motor **134** is an electric motor, these components can include a control console or a battery that sources a drive current. If the motor **134** is a pneumatic motor. The drive components are the source of pressurized gas and the line over which the gas is supplied to the saw **130**. If the motor **134** is a hydraulically driven motor, these components are the source of pressurized liquid and the line over which the liquid is supplied to the saw **130**.

(41) Saw **130**, like saw **30**, may be designed to reciprocate a conventional reciprocal saw blade.

(42) A front end attachment **150** is removably attached to the distal end of the saw body **132**. The attachment **150**, together with the saw **130** and the blade cartridge **180**, may collectively comprise a second example saw system. Front end attachment **150** has a base **152**. Base **152** is the body of the attachment **150**. Base **152** is dimensioned to seat over and be removably attached to the distal end of the saw body **132**. The base **152** may be the first coupling feature by which the base is removably attached to the saw body **132**. One possible coupling structure is complementary threading on the saw **130** and on the attachment base **152**. An alternative coupling structure may be provided by one of the saw **130** or the attachment **150** having a snap lock that engages the other of the attachment or the saw. Still another structure for removably holding attachment **150** static to the saw body is to provide one or more compressible members that provide a friction hold or a snap fit of the attachment **150** to the saw **130**.

(43) Forward of the base **152**, the front end attachment **150** has a nose **154**. The attachment **150** is formed so that as the nose **154** extends distally from the base **152**, the diameter of the nose **154** decreases. The attachment **150** is generally hollow. Not identified with reference numbers are the proximal and distal openings into the nose **154**. Attachment **150** is further formed so a slot **156** is formed in the nose **154**. Slot **156**, an example second coupling feature, is elongated in shape and has a major axis that is parallel to the longitudinal axis through the attachment **150**. The attachment **150** is formed so that the distal end of the slot **156** is spaced proximally from the distal end of the nose **154**. Forward of slot **156**, the attachment **150** may have two diametrically opposed slots. The opposed slots extend proximally from the distal end of the attachment nose **154**. One of the opposed slots is aligned with slot **156**.

(44) Cartridge **180**, as seen best in FIGS. 10 and 11 includes a static bar **182** in which a rack **212** is

removably mounted. Bar **182** is formed from two plates **184**. The plates **184** are generally rectangular in shape. More specifically, each plate **184** has a top face **190** the edge of one of which is identified in FIG. **11**. Forward of the top face **190**, each plate **184** has a distally directed face **186** the edge of one of which is identified in FIG. **11**. Between the distally directed face **186** and the top face **190** there is a transition face **188**, the edge of one of which is identified in FIG. **11**. A plate **184** is formed so that, as the transition face **186** extends proximally from the distally directed face **186** tapers upwardly to the top face **190**. The plates **184** are further formed so a rib **191** protrudes upwardly from the top faces **190**. Each rib **191** extends forward a short distance, less than 1 cm, from the proximal end of the plate **184** with which the rib is integral.

(45) Each plate **184** is formed so as to have a slot **192** that is located entirely within the plate and that is located inward of the distally directed face **190** and a short distance above the bottom face **185**. Each slot **192** has an elongated section that extends parallel to the distally directed face **186** and an elongated section that extend parallel to the transition face **188**. Each plate **184** also has a slot **194** that extends distally forward from proximal end of the plate. The plates are formed so the distal end perimeters of slots **194** are semi-circular in shape.

(46) A spacer **198** is disposed between the plates **184**. Spacer **198** includes a beam **202** similar to the first described beam **102**. Thus beam **202** has sections that are flush with the distally directed faces **186**, the transition faces **188** and the top faces **190** of the plates **184**. The portion of the beam **202** that extends flush with the plate top faces **190** does not extend to the proximal end of the top faces **190**. Instead, spacer **198** is further formed to have leg **201** that extends proximally between the proximal portions of the plate top faces **190**. Leg **201** while integral with beam **202** has a top to bottom height less than that of beam. The spacer **198** is formed so that leg **201** extends proximally rearward beyond the plates **184**. A foot **203**, an example first holding feature, projects laterally upwardly from the section of the leg **201** located proximal to the plates **184**.

(47) A single web **204** extends perpendicularly from the beam **202**. A foot **206** is located at the end of web **204**. Web **204** and foot **206** have the same relationship to each other and beam **202** as web **104** and feet **106** due to beam **102**.

(48) When cartridge **180** is assembled, beam **202**, web **204** and foot **206** are held static between the plates **184**. Leg **201** and foot **203** are able to flex up and down.

(49) Rack **212** includes a base **218**. Base **218** is analogous in shape and function to base **118** of cartridge **80**. Thus base **218** has a single T-shaped opening, i.e., a guide slot **220** in which the cartridge web **204** and foot **206** seat. A leg **217**, also part of the rack **212**, extends proximally from the base **218**. Leg **217** extends into the space between the slots **194** formed in the cartridge bar **182**. Rack **212** is further formed to have a head **224** from which a set of linearly aligned teeth **126** extend. Head **224** extends outwardly from the section of the base **218** adjacent the bottom faces of the bar plates **184**. Again it should be understood that the head **224** and teeth **126** have a thickness greater than that of the rack leg **217** and base **218** and at least as great as the side to side thickness of the cartridge bar **182**. Thus, the components of cartridge **180** are designed to that the kerf cut by the teeth is of sufficient width to accommodate the cartridge bar **182**.

(50) An arbor **228** extends proximally from the leg **217** of the rack **212**. Arbor **228** is in the form of a solid cylindrical rod. Not identified with a reference number is a slot that extends proximally from the distal end of the arbor. This slot which extends across the arbor **228**, is dimensioned to receive leg **217** integral with rack **212**. The arbor **228** is braised or welded over the leg **217**. When cartridge **180** is assembled, the distal sections of the arbor that extend outwardly of the rack **212** seat in slots **194** formed in the blade bar. The components forming the cartridge **180** are dimensioned so that the arbor **228** can reciprocate in the bar slots **194**.

(51) The proximal end of the arbor **228** is formed with retention features, i.e., example second holding features **230**. The retention features are dimensioned to engage complementary retention features integral with the reciprocating shaft internal to the saw. These features when engaged, releasably hold the arbor **228** to the reciprocating shaft so the arbor reciprocates with the shaft. The

retention features may include a set of notches **230** formed in the arbor.

(52) Saw **130** and cartridge **180** are readied for use by first fitting attachment **150** over the distal end of the saw body **132**. Cartridge **180** is then fitted to the saw. As part of this process, arbor **228** is coupled to the reciprocating shaft internal to the saw **130**. The leg **201** integral with spacer **202** is inserted into the attachment **150** until the associated foot **203** snaps into slot **156** formed in the attachment. The seating of foot **203** in the attachment slot **156** releasably holds the cartridge bar **182** to attachment **150** and, by extension, the saw **130**. As a result of the attachment of the cartridge **180** to the attachment, rib **191** and the portion of the spacer **202** sandwiched between the ribs seat in the opposed slot adjacent slot **156**. The ends of plates **184** opposed to ribs **191** seat in the second slot.

(53) The saw **130** with attached cartridge **180** is used in the same manner in which a conventional micro-style reciprocating saw is used. The actuation of the motor internal to the saw results in the back and forth movement of the reciprocating shaft. The reciprocation of the rod is, through the arbor **228** transferred to the rack **212** to cause the like movement of the rack. Owing to the dimensioning of the components, the seating of foot **206** in the rack opening **220** holds the rack for reciprocation in the cartridge bar **182**. The movement of the teeth **126** saws the tissue against which the head **224** is applied.

(54) During the cutting process, debris may enter the space between plates **184**. The debris are discharged through slots **194**.

### III. Third Embodiment

(55) FIGS. **12** and **13** depict a second alternative blade cartridge, cartridge **240** attached to the saw **130** and attachment **150**. The attachment **150**, together with the saw **130** and the blade cartridge **240**, may collectively comprise a third example saw system. Cartridge **240**, seen best in FIGS. **14** and **15**, includes a blade bar **242**. The blade bar **242** is formed from two plates **244**, seen best in FIG. **15**. Plates **244** have a distal end that is generally in the shape of the distal end of plates **184** of cartridge **180**. Plates **244** differ from plates **184** in that plates **244** have a top face **249** that extends proximally from the front face **248**, initially curves upwardly and then tapers upwardly. The tapered section of top face **249** subtends a length between 30 and 70% of the overall length of the plate **244** with which the face is integral. Plates **244** include proximally located slots **245** similar in shape and function to previously described slots **194**.

(56) Windows **246** are formed in plates **244** so as to be below and parallel to the tapered sections of plate top faces **249**.

(57) A spacer **250**, also part of the blade bar **242**, is located between the plates **244**. Spacer **250** has a foot **252** and a leg **254** similar to the foot **203** and leg **201** of cartridge **180**. A beam **256** extends distally from the leg **254**. Beam **256** is shaped to be flush with the top faces of the plates **244** between which the beam is disposed. Thus, the proximal section of beam **256** extends along a line that is parallel to the longitudinal axis of the cartridge. A distal section **258** of the beam **256**, extending distally from the proximal section, tapers downwardly. The distal end of beam **256** is formed to define a finger **260** that is angled relative to the adjacent tapered distal section **258**. Finger **260** is shaped to have an outer face, not identified, that is flush with the outer faces of the plate **244**. The beam **256** is further formed to have a second finger, finger **262**. Finger **262** projects proximally from the inner face of the beam section **258**, the face of the beam section disposed within the plates **244**. Finger **262** extends along a line that is parallel to or in registration with the longitudinal axis of the cartridge **180**.

(58) Also part of cartridge **240** is a rack **270**. Rack **270** includes a base **274** dimensioned to seat in and reciprocate in the void space between the plates **244** that is proximal to spacer **250**. The rack base **274** has a top edge, not identified that is tapered. The taper of the rack top edge corresponds to the taper of the beam distal section **258**. Rack base **274** is formed to have an elongated slot, i.e., a guide slot **276**. The slot **276** extends proximally from the tapered top edge of the rack base **274**. The rack **270** is formed so that the slot **276** is positioned to and dimensioned to receive the finger

**262** internal to the cartridge bar **242**.

(59) A head **280**, also part of the rack **270**, extends from the bottom of the rack base **274**. Head **280** is the portion of the rack **270** that is located outwardly the bottom end of the blade bar **242**. A section of the head **280** is located forward of the distal end of the rack base **274**. The head **280** is formed to have teeth **282**. The teeth are shaped to cut the tissue against which the cartridge **240** is applied. The thickness of at least the teeth is greater than the thickness across the rack base **274**. More particularly, the teeth have a thickness so that the kerf cut by the teeth can accommodate the insertion of the cartridge bar into the kerf. At a minimum, the teeth **282**, if not the whole of the head **280**, has a thickness at least equal to the side-to-side thickness between the outer faces of plates **244**.

(60) Cartridge **240** includes the previously described arbor. Arbor **228** is welded or otherwise secured to the proximal portion of the rack base **274**.

(61) Cartridge **240** is fitted to the saw **130** using the same technique employed to fit cartridge **180** to the saw. The seating of foot **252** in the slot **156** releasably holds the cartridge to the saw **130**.

(62) Cartridge **240** is used in the manner in which cartridge **130** is employed. The seating of finger **262** in rack slot **276** holds the rack to the bar **242**. Cartridge **240** has a distal section with a relative short top to bottom height. A cartridge height, from the cutting edge of the teeth **282** to the top of the blade bar of the distalmost 1 cm portion of the cartridge, may be a maximum of 1.5 cm. Such a cartridge height may facilitate the positioning of the cartridge in confined spaces such as during intra oral cutting of the mandible in orthognathic procedures or when cutting the maxilla adjacent the infraorbital nerve. Cartridge **240** is thus well suited to cut tissue in locations where it may be difficult to position a cartridge of larger bottom-to-top height.

(63) During the cutting of tissue, the debris generated by the cutting process that enter the space between the plates **240** are discharged from the bar through windows **246**.

#### IV. Alternative Embodiments

(64) Alternative versions of the claimed invention may have features different from what has been described. For example, the features of the disclosed systems and components may be combined as necessary.

(65) Likewise, the structural features may be different from what has been described. For example, the static member that extends into the slot may be a pin, inclusive of a pin-like structure. Further, such a pin or set of pins may be formed separately from the component that functions as the spacer between the bar plates. Further, the plural pins may seat in a single slot formed in the base of the rack. By extension, the slot in which this static component is disposed may be not open to the perimeter of the rack. This slot may be an elongated opening that is disposed entirely within the base of the rack.

(66) The component or components that hold the rack to the bar may be formed integrally with one or more of the plates that form the body of the bar. Thus, as a result of a stamping or machining process, one or more of the plates is formed with a raised member. This raised member is dimensioned to seat in the complementary elongated opening in the base of the rack. The static component may be formed by the selective molding or machining of the bar of the bar or rack with which the member is integral. It should likewise be appreciated that there is no requirement this static component have any curved surfaces. Thus, this component can be rectangular or square in shape.

(67) The slot may be a closed slot located completely in the rack.

(68) The complementary features of the bar and rack that hold the rack to the bar while allowing the rack to move linearly may be opposite what has been disclosed above, i.e., one or more static members may extend outwardly from one or both of the opposed sides of the base of the rack. The bar may be formed with one or more elongated openings, slots, to receive the static members. Thus, when the cartridge is assembled, the rack is disposed between the plates of the bar and the static member or members integral with the rack base are disposed in the one or more slots formed in the

bar. The static members and slots are formed so that the rack can move linearly in the bar. Since the static member or members are seated in the bar, the member or members hold the rack in the bar. (69) It should furthermore be understood that the static member, regardless of if this member is part of the bar or the rack, may be integrally formed with the component from which the member extends. The static member may be formed punching out a section of bar or rack with which the member is integral.

(70) The rack base may subtend a surface area that is more than 50% of the surface area of the bar in which the base is seated. The rack base may alternatively subtend an area less than this percentage of the surface of the bar.

(71) It may not be necessary to provide the bars with openings through which the debris generated in the cutting process can be discharged.

(72) The features integral with the rack to releasably hold the rack to the reciprocating shaft employed to reciprocate the rack may be different from what has been described. Likewise, the features integral with bar that facilitates holding the bar static to the associated attachment or saw to which the cartridge is coupled may vary from what has been described.

(73) The rack may be formed so that head and teeth project beyond the distal proximal end of the bar.

(74) Still another alternative blade cartridge **292** is now described with reference to FIGS. **16** and **17**. Cartridge **292** includes the previously described rack **212** and arbor **228**. Rack **212** is enclosed in a blade bar **294**. Blade bar **294** is formed to have two opposed plates **296**. Plates **296** are formed to have outer surfaces that, extending away from the side adjacent which the head **224** and teeth extend away from the bar, taper outwardly from the rack **212**. Spacer **202** separates the plates **296** so as to define the void space internal to the bar **294** in which the base **218** of the rack **212** is seated. The blade bar **294** is further formed so at the distal end to have a head **298**. Head **298** projects outwardly from the opposed outer surfaces of the plates **296**. The blade bar **294** is further formed so that this head is located distal to the distalmost location along the bar **294** from which teeth **126** extend when rack **212** reciprocates back and forth. Cartridge **292** can be used to cut bone when it is desirable limit the depth to which the cartridge is inserted by pressing a stop against the underside of the bone being cut. One such procedure is a procedure in which it is desirable to cut across the sternum without cutting the tissue below the sternum. In this type of procedure, the protruding head **298** of the cartridge bar performs the same function as the foot of a sternum guard. (75) A benefit of cartridge **292**, sometimes called a sternum cartridge, is that this cartridge can be fitted to a saw **30** or attachment to which any of the above disclosed cartridges **80**, **180** or **240** may be attached. This means it is possible to use a cartridge to cut through the sternum without going to the expense of providing a saw especially designed to hold a sternum guard. The tapered surfaces of the bar **294** serve to spread the sternum after the teeth **126** form the initial cut.

(76) Another procedure in which the cartridge **292** is useful is when the cartridge **292** is used to make a cut through the skull. In this type of procedure, the stop integral with the cartridge bar is employed to ensure the cartridge is not inserted to a depth that will result in the unintended cutting of tissue below the skull.

(77) From the above it should be understood that there is no requirement that the bar be constructed so the opposed sides are planar and parallel.

(78) It should be appreciated that when the bar has a tapered shape the head and teeth of the rack may not have a width thereacross equal to the maximum width of the bar. Instead, the head and teeth may be formed so that the kerf cut by the rack is at a minimum of sufficient depth to receive the adjacent section of the bar from which the rack head extends. Here the adjacent section of the bar is generally understood to be the first 3 to 10 mm portion of the bar that follows the head of the rack into the bone and, typically, no more than the first 8 mm portion of the bar that follows the head of rack into the bone. The wedge forming portion or portions of the bar that follow this bottom section are understood to, after the bone is cut, further separate the cut sections of bone

apart from each other.

(79) It should further be understood that in alternative system, the saw may include features for both releasably holding the cartridge bar static to the body of the saw and releasably holding the rack to the reciprocating shaft. Thus, the features of the attachments that hold the bar static to the attachment may simply be incorporated into the saw. This saw is of use if the facility determines it is in the interest of the facility to simply have a saw available that, without an attachment, can be used to actuate the blade cartridge **80**.

(80) Accordingly, it is an object of the appended claims to cover all such variations and modifications that come within the true spirit and scope of this disclosure.

## Claims

1. A surgical saw blade assembly for use with a surgical saw, the surgical saw blade assembly comprising: a blade cartridge including: a bar including a first holding feature; and a rack, the rack moveably disposed within the bar and configured to move along a proximal-to-distal longitudinal axis of the bar and is in removable engagement with a surgical saw; and an attachment disposed between and connecting the bar of the blade cartridge to the surgical saw, the attachment including: a coupling feature in selective releasable engagement with the first holding feature of the bar such that the coupling feature engages the first holding feature to hold the bar static when the rack is reciprocated along the proximal-to-distal longitudinal axis of the bar.
2. The surgical saw blade assembly of claim 1, wherein the coupling feature of the attachment defines an opening dimensioned to receive the first holding feature of the bar.
3. The surgical saw blade assembly of claim 1, wherein the coupling feature of the attachment is a finger that is dimensioned to seat in an opening defined in the bar of the blade cartridge, the opening being the first holding feature of the blade cartridge.
4. The surgical saw blade assembly of claim 1, wherein the attachment further comprises a body including a first coupling feature to releasably hold the body of the attachment to the saw.
5. The surgical saw blade assembly of claim 4, wherein the attachment includes two shells that form the body of the attachment, each of the shells includes a planar base and two wings that extend outwardly from opposed sides of the planar base.
6. The surgical saw blade assembly of claim 5, wherein each of the shells is shaped so that as the wings extends away from the planar base of the shell, the wings curve away from the planar base; and wherein, when assembled, ends of the each of the wings of each of the shells face the ends of the wings of the other shell.
7. The surgical saw blade assembly of claim 5, wherein the attachment further comprises a hand that extends distally forward from each shell, each hand includes a rectangular palm; and wherein each of the palms are dimensioned to seat over an exposed surface of the shell from which the hand extends.
8. The surgical saw blade assembly of claim 7, wherein each of the hands comprises a finger that extends forward from each palm, each of the fingers has a main section that extends along a longitudinal axis that is parallel to a proximal-to-distal longitudinal axis through the saw.
9. The surgical saw blade assembly of claim 8, wherein each of the fingers includes a tip that extends perpendicularly to the longitudinal axis of a main section of the finger; and a rib, that defines the coupling feature, protrudes forward from an end face of the tip of each of the fingers and the fingers are arranged so that opposed ribs of each of the fingers face each other and engage the first holding feature of the bar to hold the bar static relative to the saw.
10. The surgical saw blade assembly of claim 4, wherein the first coupling feature includes components attached to the body to clamp the body to the surgical saw.
11. The surgical saw blade assembly of claim 1, wherein the rack of the blade cartridge further comprises: a base; and wherein one of the base and the bar includes an opening; and wherein the

other of the base and the bar includes a static member extending into the opening so as to retain the rack in the bar, and the opening and the static member are collectively shaped to allow the rack to engage in reciprocal movement from proximal to distal to proximal along a longitudinal axis of the bar.

12. A surgical saw blade assembly for use with a surgical saw, the surgical saw blade assembly comprising: a blade cartridge including a bar and a rack at least partially disposed within the bar, the bar including a first holding feature; an attachment connecting the blade cartridge to a surgical saw, the attachment for releasably holding the bar of the blade cartridge static relative to the surgical saw when the rack is reciprocated along a proximal-to-distal longitudinal axis of the bar; a drive link disposed between and connecting the surgical saw to the rack of the blade cartridge, the drive link arranged to translate the reciprocal motion of the surgical saw to the rack of the blade cartridge.

13. The surgical saw blade assembly of claim 12, wherein the rack of the blade cartridge further comprises: a base; and wherein one of the base and the bar includes an opening; and wherein the other of the base and the bar includes a static member extending into the opening so as to retain the rack in the bar, and the opening and the static member are collectively shaped to allow the rack to engage in reciprocal movement from proximal to distal to proximal along a longitudinal axis of the bar.

14. The surgical saw blade assembly of claim 12, wherein the attachment comprises a coupling feature, the coupling feature of the attachment defines an opening dimensioned to receive first holding feature of the bar.

15. The surgical saw blade assembly of claim 12, wherein the attachment comprises a coupling feature, the coupling feature is a finger that is dimensioned to seat in an opening defined in the bar of the blade cartridge, the opening being the first holding feature of the blade cartridge.

16. The surgical saw blade assembly of claim 12, wherein the attachment further comprises a body including a first coupling feature in releasable engagement with a distal end of the surgical saw to releasably hold the body of the attachment to the surgical saw.

17. The surgical saw blade assembly of claim 12, wherein the attachment includes two shells that form a body of the attachment, each of the shells includes a planar base and two wings that extend outwardly from opposed sides of the planar base.

18. A surgical saw blade assembly for use with a surgical saw, the surgical saw blade assembly comprising: a blade cartridge including a bar and a rack wherein the rack is moveably disposed within the bar, and the rack is in removable engagement with a surgical saw, and the bar includes a first holding feature; and an attachment disposed between and connecting the bar and the surgical saw, the attachment including: a body including a first coupling feature in releasable engagement with a distal end of a housing to releasably hold the body to the saw; a second coupling feature in selective releasable engagement with the first holding feature of the bar; and wherein the second coupling feature of the attachment has an opening dimensioned to receive the first holding feature of the bar.

19. The surgical saw blade assembly of claim 18, wherein the rack of the blade cartridge further comprises: a base; and wherein one of the base and the bar includes an opening; and wherein the other of the base and the bar includes a static member extending into the opening so as to retain the rack in the bar, and the opening and the static member are collectively shaped to allow the rack to engage in reciprocal movement from proximal to distal to proximal along a longitudinal axis of the bar.

20. The surgical saw blade assembly of claim 18, further comprising a drive link disposed between and connecting the surgical saw to the rack of the blade cartridge, the drive link arranged to translate the reciprocal motion of the surgical saw to the rack of the blade cartridge.

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